

MACWORLD · SAN FRANCISCO · 9 JANUARY 2007

Reinventing the phone

How one keynote turned the mobile phone into a pocket computer

ACHIM KIPKORIR LANGAT · COMPUTER SCIENCE

IN THIS PRESENTATION

Contents

Five chapters on the device that turned phones into computers.

01 **Context**
What smartphones were like before 2007

02 **The keynote**
The presenter, the announcement, and the demo behind it

03 **The engineering**
Hardware, software stack, multi-touch and sensors

04 **The platform**
Launch, reception, the SDK and the App Store

05 **The impact**
What changed for computing, and lessons for engineers

01 · CONTEXT

Smartphones before 2007

Capable devices, built around fixed hardware controls and a cut-down version of the web.

01

Plastic keyboards

Physical keys filled half the front and stayed the same whether an app needed them or not.

02

Styluses

Many touchscreens were pressure-based and needed a pen for accurate taps. Fingers were an afterthought.

03

The “baby internet”

Jobs' phrase for the stripped-down mobile versions of websites that most phones displayed.

04

Buried features

Deep menu trees hid what phones could do. Jobs' complaint: they were neither very smart nor easy to use.

Leading platforms in 2006: Symbian (Nokia), BlackBerry, Windows Mobile, Palm OS

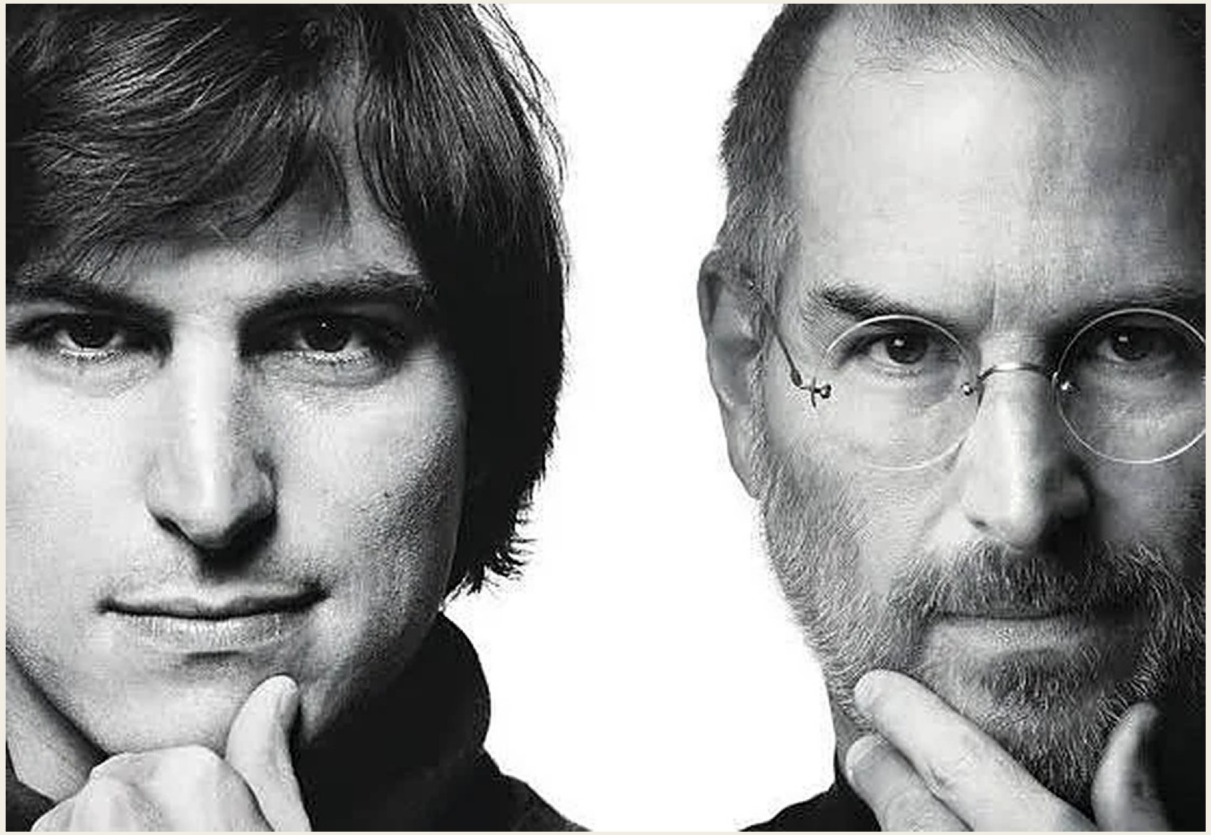


Fig. 1 · Steve Jobs as a young founder and in his later years as Apple's chief executive.

02 · THE KEYNOTE

The man on stage

1976	Co-founds Apple with Steve Wozniak, aged 21.
1985	Pushed out of Apple; starts NeXT and buys the studio that becomes Pixar.
1997	Returns after Apple buys NeXT. NeXT's operating system becomes Mac OS X.
2007	Unveils the iPhone at Macworld. He died in October 2011, aged 56.

The announcement



Fig. 2 · Jobs holds up the iPhone at Macworld, 9 January 2007.

Jobs promised three revolutionary products:

1. **A widescreen iPod with touch controls**
2. **A revolutionary mobile phone**
3. **A breakthrough internet communications device**

Then the reveal: they were not three devices. They were one.

Source: Apple press release, 9 January 2007



MACWORLD KEYNOTE · 9 JANUARY 2007

“Today, Apple is going to reinvent the phone.”

STEVE JOBS

02 · THE KEYNOTE

Behind the demo

01 **The golden path**
One rehearsed order of actions that the prototype could survive.

02 **Several phones on stage**
Jobs switched between units because each could run out of memory.

03 **A portable cell tower**
AT&T brought one in so the live call would have a strong signal.

04 **Five bars, always**
Demo units were hard-coded to show full signal strength.

THE PROTOTYPE ON DEMO DAY

- Could play part of a song or video, but not a full clip reliably
- Often ran out of memory after a handful of tasks
- Radio and memory software still had bugs

Lesson: test the paths real users take, not only the rehearsed one.

Source: Fred Vogelstein, New York Times Magazine, 2013

The hardware



Fig. 3 · Launch image: 9:41 on Tuesday, January 9.

Component	Original iPhone (2007)
Processor	Samsung 32-bit ARM11, rated 620 MHz, run at 412 MHz
Graphics	PowerVR MBX Lite GPU
Memory	128 MB RAM
Storage	4 GB or 8 GB flash (16 GB added in 2008)
Display	3.5-inch, 320 × 480 (163 ppi), capacitive multi-touch
Networks	Quad-band GSM, EDGE, Wi-Fi 802.11b/g, Bluetooth 2.0
Sensors	Accelerometer, proximity, ambient light
Camera	2 megapixels, still photos only

DESIGN TRADE-OFF

Why run a chip rated 620 MHz at 412 MHz? Dynamic power, $P \approx C \cdot V^2 \cdot f$, grows with clock speed and the square of voltage, so slowing down cut heat and stretched battery life.

Sources: Apple, EveryMac, Low End Mac

The software stack

Apps: Phone · Mail · Safari · iPod · Maps · Photos

Cocoa Touch
Touch events, UI controls, views, accelerometer

Media
Core Audio, OpenGL ES, Core Animation, video

Core Services
SQLite, networking, Core Location, contacts

Core OS
OS X kernel, BSD sockets, power, security

Runs OS X

Built on the Mac's kernel and frameworks, not a phone OS written from scratch.

A real browser

Safari on the WebKit engine showed full desktop web pages, zoomed with a double tap or a pinch.

Desktop-class apps

Mail with IMAP and POP, Google Maps, and visual voicemail you could play in any order.

Under half a gigabyte

The entire operating system used less than 0.5 GB of flash storage.

Layer names as Apple published them with the iPhone SDK, March 2008

Multi-touch input

1
Sense
A grid of capacitive sensors under the glass registers each finger

→

2
Locate
Software separates the readings into individual touch points

→

3
Interpret
A gesture recognizer tracks the points over time: tap, flick or pinch

→

4
Respond
The app receives a touch event and updates the screen

GESTURES IT MADE FAMILIAR

Tap
Open, select and type

Flick
Scroll lists with momentum

Pinch
Zoom photos, maps and pages

WHERE IT CAME FROM

In 2005 Apple acquired FingerWorks, founded by University of Delaware researchers Wayne Westerman and John Elias, makers of multi-finger gesture trackpads.

Sources: Purdue ECE (Westerman profile), Apple press release 2007

Context-aware sensors

Three small sensors let software adapt to how the phone is held, with no extra buttons.

Accelerometer

DETECTS

Which way the phone is tilted and how it moves

SOFTWARE RESPONSE

Rotates photos, web pages and the keyboard between portrait and landscape

Proximity sensor

DETECTS

Your face close to the screen during a call

SOFTWARE RESPONSE

Turns off the display and touch input, preventing cheek taps and saving power

Ambient light sensor

DETECTS

How bright the surroundings are

SOFTWARE RESPONSE

Adjusts screen brightness to suit the room and save battery

CS idea: *context-aware computing. The device adapts without being asked.*

Source: Apple press release, 9 January 2007

Launch and early sales

June 2007

On sale in the US on 29 June, only through AT&T

$499

for 4 GB, or $599 for 8 GB, with a two-year contract

74 days

to sell the first million. The iPod had taken almost two years.

6.1M

original iPhones sold before the iPhone 3G replaced it in July 2008

Sources: Apple press releases 2007; Low End Mac

Version-one limits and doubters

WHAT VERSION ONE COULD NOT DO

- No third-party apps; developers were told to build web apps
- No 3G: data ran over the slower EDGE network
- No copy and paste
- No picture messaging (MMS)
- Camera took photos only, no video
- Locked to one carrier in each country

NOT EVERYONE WAS CONVINCED

> *"There's no chance that the iPhone is going to get any significant market share."*

Steve Ballmer, Microsoft CEO, USA Today interview, April 2007

Apple's own public target: 10 million iPhones sold in 2008.

Sources: The Register, 2 May 2007; Mac Daily News, 10 September 2007

The platform shift

6 MARCH 2008

iPhone SDK

- Xcode, Interface Builder and an iPhone Simulator
- Native apps in Objective-C, built on Cocoa Touch
- Access to the same four framework layers shown earlier

10 JULY 2008

App Store opens

- More than 500 native apps on day one, over 125 of them free
- 10 million downloads in the first weekend
- Developers keep 70% of each sale; Apple takes 30%

Why it mattered: software no longer had to go through carriers or shop shelves. A small developer could reach every iPhone owner directly.

Sources: Apple press release, 10 July 2008; Macworld, 14 July 2008

Timeline

2005
Apple buys FingerWorks; phone work under way

Jan 2007
Announced at Macworld

Jun 2007
On sale in the US

Sep 2007
One millionth iPhone sold

Mar 2008
iPhone SDK released

Jul 2008
App Store and iPhone 3G launch

Jul 2016
One billionth iPhone sold

Jul 2025
Over 3 billion sold

Nine years to the first billion iPhones; nine more to pass three billion.

Sources: Apple; Loop Insight, 2016; GSMArena, 2025

Impact on computing

01

Touch-first interaction

Direct manipulation with fingers and gestures became the default way most people use a computer.

02

A mobile-first web

Full browsers on phones pushed websites toward responsive, touch-friendly design.

03

App store distribution

One-tap installs and automatic updates made app stores the main channel for consumer software.

04

Low-power ARM chips

Efficiency beat raw clock speed. Apple went on to design its own chips, starting with the A4 in 2010.

05

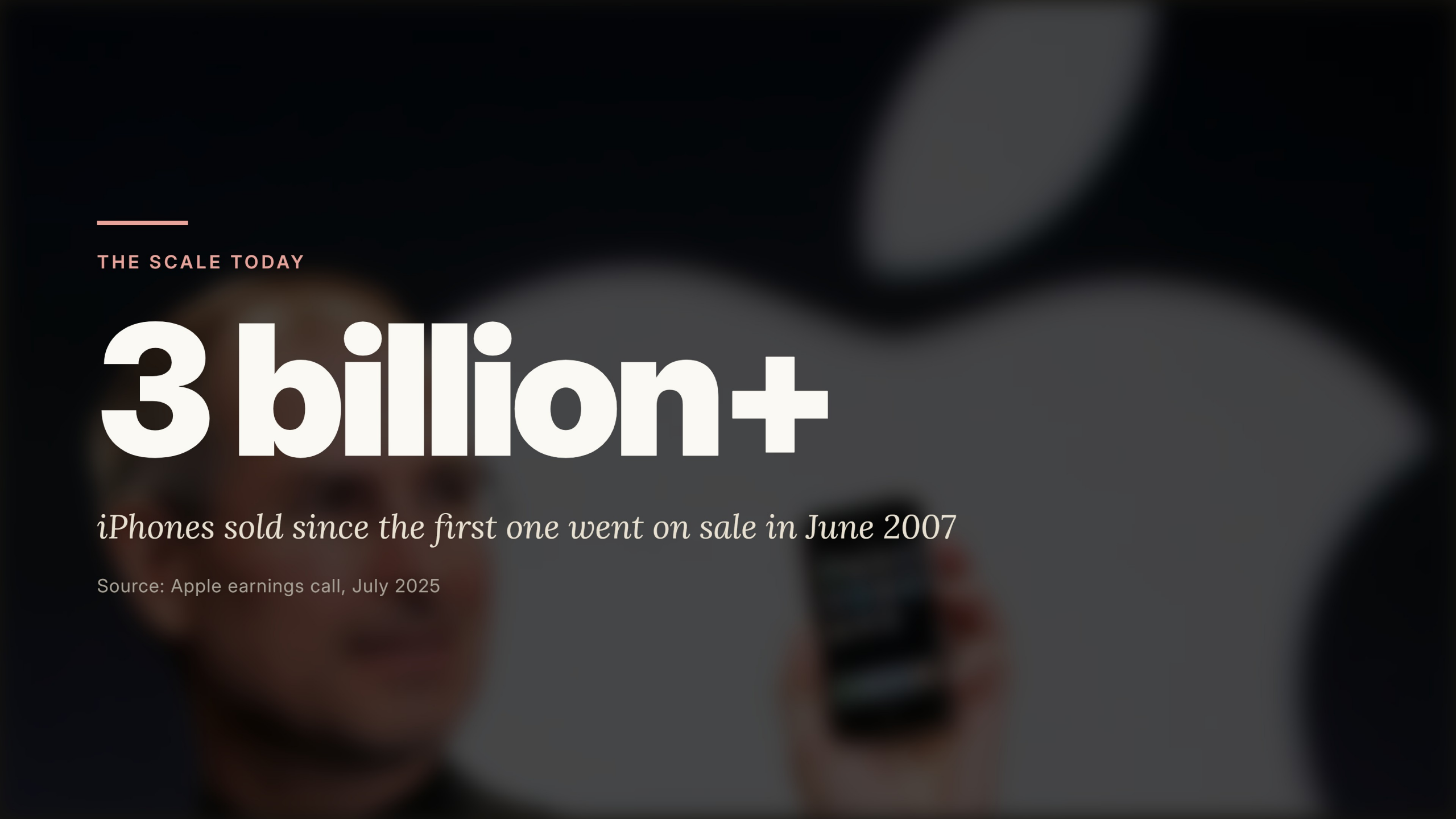
Location-aware services

GPS, added in 2008, plus always-on data made maps, ride-hailing and delivery apps possible.

06

Cloud-connected devices

Phones became the front end for cloud services: sync, push notifications and streaming.



THE SCALE TODAY

3 billion+

iPhones sold since the first one went on sale in June 2007

Source: Apple earnings call, July 2025

05 · THE IMPACT

Lessons for engineers

01

Integrate the whole system

Hardware, operating system and apps were designed together, so a 412 MHz chip could still feel fast and fluid.

02

Turn fixed hardware into software

One button and an on-screen keyboard let the interface change shape for every app.

03

Ship version one, then open the platform

The SDK and App Store arrived a year after launch and multiplied what the phone could do.

04

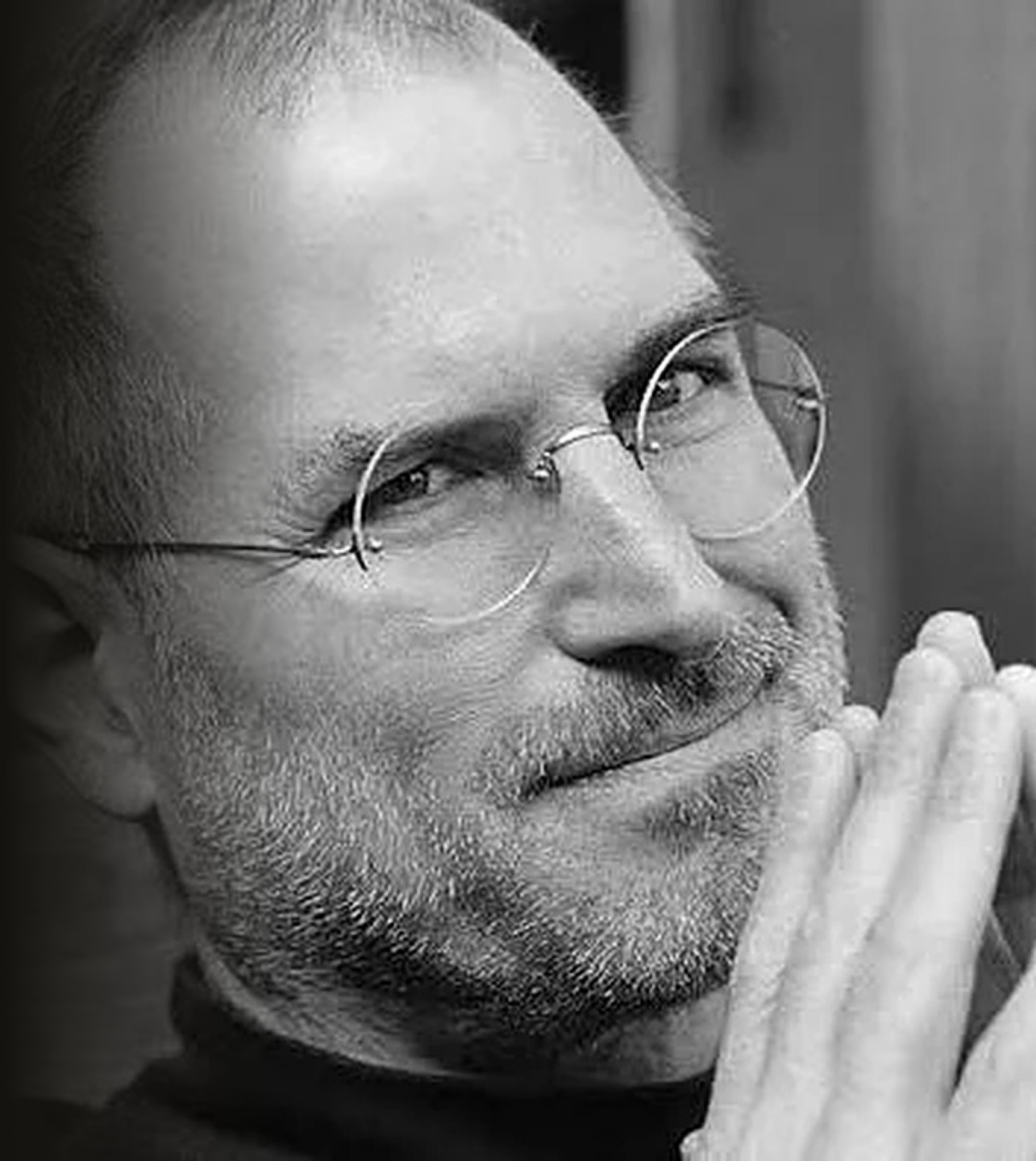
A demo is not a product

The golden path carried the keynote; months more engineering made the phone work for real users.

Thank you.

Questions?

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BACK MATTER

Sources and credits

SOURCES

- Apple. Apple Reinvents the Phone with iPhone. Press release, 9 January 2007.
- Apple. Apple Sells One Millionth iPhone. Press release, 10 September 2007.
- Apple. iPhone 3G on Sale Tomorrow. Press release, 10 July 2008.
- Vogelstein, F. And Then Steve Said, "Let There Be an iPhone". New York Times Magazine, October 2013 (via Gizmodo).
- EveryMac: Apple iPhone (Original) specs.
- Low End Mac: Original Apple iPhone.

SOURCES, CONTINUED

- Purdue ECE: Dr. Wayne C. Westerman.
- iClarified: iPhone SDK Roadmap Event Summary, 6 March 2008.
- Macworld: 10 million apps, 1 million iPhone 3Gs, 14 July 2008.
- The Register: Ballmer says iPhone will be a niche player, 2 May 2007.
- GSMArena: Apple has sold more than 3 billion iPhones, 1 August 2025.
- Jobs, S. Macworld San Francisco keynote, 9 January 2007 (video).

PHOTOGRAPHS

- **Slides 1, 17:** Jobs with the iPhone, Macworld 2007. [Photographer / agency]
- **Slide 4:** Jobs, young and in later life. [Photographer]
- **Slides 5, 6:** Jobs on stage, Macworld 2007. [Photographer / agency]
- **Slide 8:** iPhone launch image. Apple, 2007.
- **Slide 19:** Portrait of Jobs. [Photographer]